

Q1.

2. SQL

< 1 2 3 4 5 6 7 8 9 10 >

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Question 1

Revisit Later

What will be the output of the following SQL query?

```
1 SELECT emp_name
2 FROM employees
3 WHERE emp_name LIKE '%a%';
4
```

Select the correct option from the given choices.

←

→

Select an option

☐ Names starting with the alphabet 'a'

☐ Names having the alphabet 'a' at the end

☐ Names having the alphabet 'a' in any position

☐ Names having the alphabet 'a' in between

Correct Answer: Names having the alphabet 'a' in any position

Explanation: The LIKE '%a%' pattern uses % as a wildcard before and after a. Therefore, the name can have any characters before or after a, meaning a can appear **in any position** within the name.

Q2.



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MCQ 

Unanswered

 Revisit Later

Question 2

Which of the following statements will you use to retrieve items in an ascending order from a table in SQL?



Select an option  Clear Response

☐ Select * from [table] order by [tbl_clm];

☐ Select * from [table] sort by [tbl_clm];

☐ Select * from [table] order by [tbl_clm] asc;

☐ Select * from [table] group by [tbl_clm] asc;

Correct Answer: SELECT * FROM [table] ORDER BY [tbl_clm] ASC;

Explanation: The ORDER BY clause is used to sort records. The ASC keyword specifies **ascending order**, so this query retrieves the table records sorted from the lowest value to the highest value of the specified column.

Q3.



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Question 3

What will be the output of the following pseudo code?

```
1 Integer p,q,r
2 Set p=1, q=8, r=9
3 q=(11+r)&q
4 p=q+q
5 Print p+q+r
```

Select the correct option from the given choices.

Select an option

☐ 6

☐ 9

☐ 13

☐ 24

Correct Answer: 9

Explanation: Initially, $p = 1$, $q = 8$, and $r = 9$. First, calculate $q = (11 + r) \& q = (11 + 9) \& 8 = 20 \& 8 = 0$, because the bitwise AND of 20 (10100) and 8 (01000) is 0. Then $p = q + q = 0 + 0 = 0$. Finally, $p + q + r = 0 + 0 + 9 = 9$.

Q4.

accenture

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2. Pseudo Code

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Question 4

Revisit Later

What is the final value of 'total' after executing the following pseudocode?

```
1 FUNCTION incrementByThree(n)
2   RETURN n + 3
3 END FUNCTION
4
5 total = 0
6 FOR i = 1 TO 3
7   total = total + incrementByThree(i)
8 END FOR
9 DISPLAY total
```

Select the correct answer from the given choices.

<

Select an option

Clear Response

☐ 15

☐ 12

☐ 18

☐ 9

>

Correct Answer: 15

Explanation: The function `incrementByThree(n)` returns $n + 3$. The loop runs for $i = 1$ to 3, producing 4, 5, and 6. Therefore, $\text{total} = 4 + 5 + 6 = 15$.

Q5.

accenture

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2. Pseudo Code

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MCQ Unanswered

Revisit Later

Question 5

What does the following pseudocode accomplish?

```
1 FUNCTION cal(number):
2   IF number is 0 OR number is 1
3     RETURN 1
4   ELSE:
5     RETURN some_operation(number, cal(number - 1))
6   End IF
7 END FUNCTION cal
8
9 result = cal(input_number)
10 PRINT result
```

←

Select an option

Clear Response

☐ Calculates the factorial of `input_number`.

☐ Computes the sum of `input_number` and its predecessors.

☐ Multiplies `input_number` by the product of its predecessors.

☐ Determines the square of `input_number`.

Correct Answer: Calculates the factorial of input_number

Explanation: The function uses recursion by calling itself with number - 1. When number reaches 0 or 1, it returns 1. Multiplying each number by the result of its predecessor produces the factorial, i.e., $n! = n \times (n-1) \times \dots \times 1$.

Q6.

2. Pseudo Code

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Question 5

Revisit Later

What will be the output of the following pseudo code for a=2, b=8, c=8?

```
1 Integer funn(Integer a, Integer b, Integer c)
2   c=a+a
3   if((4+10*a)<(7+c))
4       if((7+a)>c)
5           if((5*b)<(b+5))
6               b=a+b
7           End if
8       End if
9   End if
10  Print a+b+c
```

Select the correct option from the given choices.

←

→

Select an option

Clear Response

☐ 17

☐ 22

☐ 21

☐ 29

Correct Answer: 21

Explanation: Initially, $a = 2$, $b = 8$, and $c = 8$. First, $c = 9 + a = 9 + 2 = 11$. Now check the condition $(4 + 10*a) < (7 + c)$, which becomes $(4 + 20) < (7 + 11)$, or $24 < 18$, which is **false**. Therefore, none of the nested if statements execute, and b remains 8. Finally, $a + b + c = 2 + 8 + 11 = 21$.

Q7.

2. Pseudo Code

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Question 1

Revisit Later

Assume you need to update the status of orders that were placed before January 1, 2021 to 'Shipped'.

The "orders" table has the columns: "order_id", "order_date", "customer_id", and "status".

Considering this context, which SQL query correctly performs this update? Analyze the given choices and select the correct option.

Select an option

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☐ UPDATE orders SET status = 'Delivered' WHERE order_date < '2021-01-01';

☐ UPDATE orders SET status = 'Shipped' WHERE order_date > '2021-01-01';

☐ UPDATE orders SET status = 'Shipped' WHERE status = '2021-01-01';

☐ UPDATE orders SET status = 'Shipped' WHERE order_date < '2021-01-01';

Correct Answer: UPDATE orders SET status = 'Shipped' WHERE order_date < '2021-01-01'; (Option 4)

Explanation: To change the status to 'Shipped' for records prior to January 1, 2021, the SET clause must assign 'Shipped' and the WHERE clause must check order_date < '2021-01-01'

Q8.

2. Pseudo Code ⓘ

< 1 2 3 4 5 6 7 8 9 10

Attempted: 2/10

⌵

Question 10

What will be the output of the following pseudo code?

```
1 Integer j
2 Integer arr[2][2] = {{0, 1}, {4, 1}}
3 if((3&arr[1][1]) < (arr[1][0] + 3))
4     arr[1][0] = (4 + 1) + arr[0][1]
5 else
6     arr[1][0] = 2 + arr[1][0]
7 arr[0][0] = (arr[0][0] ^ 1) + arr[0][0]
8 Print arr[1][0] + arr[0][0]/code
```

Select the correct option from the given choices.



Select an option

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☐ 7

☐ 12

☐ 9

☐ 19

Correct Answer: 7

Explanation:

1. Initial 2D Array:

$arr[0][0] = 0, arr[0][1] = 1$

$arr[1][0] = 4, arr[1][1] = 1$

2. Line 3: if $((3 \& arr[1][1]) < (arr[1][0] + 3))$

- $(3 \& 1) = 1$

- $(4 + 3) = 7$

- $1 < 7$ is True.

3. Line 4: $arr[1][0] = (4 + 1) + arr[0][1]$

- $arr[1][0] = 5 + 1 = 6$

4. Line 7: $arr[0][0] = (arr[0][0] ^ 1) + arr[0][0]$

- $(0 ^ 1) + 0 = 1 + 0 = 1$

- $arr[0][0] = 1$

5. Line 8: Print $arr[1][0] + arr[0][0]$

- Output = $6 + 1 = 7$

Q9.

2. Pseudo Code ⓘ

< 1 2 3 4 5 6 **7** 8 9 10 > ⚙

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Question 7 [Revisit Later](#)

What will be the output of the given pseudo code?

```
1 Integer j
2 Integer arr[2][2] = {{1, 2}, {2, 4}}
3 arr[0][1] = (arr[1][0] + arr[0][0]) & arr[1][1]
4 if((arr[0][0] - arr[1][1]) > (arr[1][1] + arr[0][0]))
5     arr[0][0] = (arr[1][1] + arr[1][1]) + arr[0][0]
6 End if
7 arr[0][0] = (3 + 7) & arr[0][0]
8 Print arr[1][1] + arr[0][0]
```

Note: & bitwise AND - The bitwise AND operator (&) compares each bit of the first operand to the corresponding bit of the second operand. If both bits are 1, the corresponding result bit is set to 1. Otherwise, the corresponding result bit is set to 0.

Select the correct option from the given choices.

[←](#)

Select an option [Clear Response](#)

☐ 1.0

☐ 4.0

☐ 8.0

☐ 17.0

[→](#)

Correct Answer: 4.0

Explanation: Initially, $arr = \{\{1, 2\}, \{2, 4\}\}$. First, $arr[0][1] = (arr[1][0] + arr[0][0]) \& arr[1][1] = (2+1) \& 4 = 3 \& 4 = 0$. The if condition is false, so $arr[0][0]$ remains 1. Then $arr[0][0] = (3+7) \& 1 = 10 \& 1 = 0$. Finally, $arr[1][1] + arr[0][0] = 4 + 0 = 4$

Q10.

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2. Pseudo Code

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Question 2

Revisit Later

Which of the following shows the correct usage of Group by clause?

←

Select an option

Clear Response

☐1 SELECT column_name1, column_name2
2 FROM table_name
3 WHERE column_name1 = operator value
4 GROUP BY column_name1

☐1 SELECT * FROM table_name
2 WHERE column_name = operator value
3 GROUP BY column_name

☐1 SELECT column_name, aggregate_function(column_name)
2 FROM table_name
3 WHERE column_name operator value
4 GROUP BY column_name

☐1 SELECT * FROM table_name
2 GROUP BY column_name

Correct Answer: Option C — SELECT column_name, aggregate_function(column_name) FROM table_name WHERE column_name operator value GROUP BY column_name

Explanation: The GROUP BY clause is used to group rows having the same value in a specified column, usually along with an **aggregate function** such as COUNT(), SUM(), AVG(), MAX(), or MIN(). The WHERE clause filters the rows before they are grouped. Therefore, **Option C** shows the proper usage of the GROUP BY clause.

Q11.

2. Pseudo Code ⓘ

< 1 2 3 4 5 **6** 7 8 9 10 >

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Question 6 [Revisit Later](#)

What will be the output of the following pseudo code for a=7, b=4, c=10?

```
1 Integer funn(Integer a, Integer b, Integer c)
2   if((b+c+a)<(c+a+b))
3     b=b+c
4   Else
5     if(3+c)<(a+3))
6       a=(4+5)+b
7     End if
8     c=c+a
9   End if
10  c=(b+b)+a
11  Print a+b+c
```

Select the correct option from the given choices.

[←](#)

Select an option [Clear Response](#)

☐ 24

☐ 30

☐ 26

☐ 32

Correct Answer: 26

Explanation:

- Initial values: $a = 7$, $b = 4$, $c = 10$
- Line 2: if $((b + c + a) < (c + a + b))$
 - $(4 + 10 + 7) < (10 + 7 + 4) \Rightarrow 21 < 21$ is False.
 - Jump to the Else block (line 4).
- Line 5: if $((3 + c) < (a + 3))$
 - $(3 + 10) < (7 + 3) \Rightarrow 13 < 10$ is False.
 - Line 6 is skipped.
- Line 8: $c = c + a$
 - $c = 10 + 7 = 17$
- Line 10: $c = (b + b) + a$
 - $c = (4 + 4) + 7 = 8 + 7 = 15$
- Line 11: Print $a + b + c$
 - Output = $7 + 4 + 15 = 26$

Q12.

2. Pseudo Code ⓘ

< 1 2 3 4 5 6 7 **8** 9 10 >

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Attempted: 2/10

Question 8 [Revisit Later](#)

What will be the output of the following pseudo code for a=1, b=2, c=10?

```
1 Integer funn(Integer a, Integer b, Integer c)
2   if((a+b)+(b+c)>(a+b+c))
3     if(a>b)
4       b=b+c
5       if((c+a+b)<(a+b+c))
6         c=(b+c)+c
7       End if
8   End if
9   Print a+b+c
10
```

Select the correct option from the given choices.

[←](#)

Select an option [Clear Response](#)

☐ 9

☐ 15

☐ 13

☐ 19

Correct Answer: 26

Explanation:

1. Initial values:

a = 1, b = 2, c = 10

2. Line 2: if $((a + b) + (b + c) > (a + b + c))$

- Left side: $(1 + 2) + (2 + 10) = 3 + 12 = 15$

- Right side: $(1 + 2 + 10) = 13$

- $15 > 13$ is True.

3. Line 3: if $(a > b)$

- $1 > 2$ is False.

- The inner block (lines 4 to 7) is completely skipped.

4. Line 9: Print $a + b + c$

- Values remain unchanged: a = 1, b = 2, c = 10

- Output = $1 + 2 + 10 = 13$

Q13. Assume that you want to find the count of the total occurrences of a particular character in the FirstandLastName field. Which of the SQL queries is correctly related to the given case? Note: Suppose the character is 'm' and you need to fetch it from StudentDetails.

A.

```
FROM StudentDetails
SELECT FirstandLastName,
COUNT(FirstandLastName) - COUNT(REPLACE(FirstandLastName, 'm', ''));
```

B.

```
SELECT FirstandLastName,
COUNT(FirstandLastName) - COUNT(REPLACE(FirstandLastName, 'm', ''))
FROM StudentDetails;
```

C.

```
FROM StudentDetails
SELECT FirstandLastName,
LENGTH(FirstandLastName) - LENGTH(REPLACE(FirstandLastName, 'm', ''));
```

D.

```
SELECT FirstandLastName,
LENGTH(FirstandLastName) - LENGTH(REPLACE(FirstandLastName, 'm', ''))
FROM StudentDetails;
```

Correct answer : D. SELECT FirstandLastName, LENGTH(FirstandLastName) - LENGTH(REPLACE(FirstandLastName, 'm', '')) FROM StudentDetails;

Explanation :

- Standard SQL syntax requires the **SELECT** clause to appear before the **FROM** clause.
- To count occurrences of a specific character within a string:
 1. **LENGTH(FirstandLastName)** returns the total character length of the original string.
 2. **REPLACE(FirstandLastName, 'm', '')** removes all occurrences of 'm'.
 3. **LENGTH(REPLACE(...))** returns the length of the string without the character 'm'.
 4. Subtracting the modified string length from the original string length gives the exact number of times 'm' appears.
- **COUNT()** is an aggregate function used to count rows/records, not characters in a single string, making options A and B incorrect.

Q14. Assume that you want to fetch student names and mark records. In this case, you also want to display the student's details even if the marks record is absent. Which SQL query can be used to complete the desired task? Analyze the given choices and select the correct option.

A. `SELECT E.StudentName FROM M.Mark WHERE StudentDetails E LEFT JOIN GROUP BY StudentMark M IN E.StudentId = M.StudentId;`

B. `SELECT E.StudentName, M.Mark FROM StudentDetails E LEFT JOIN GROUP BY StudentMark M IN E.StudentId = M.StudentId;`

C. `SELECT E.StudentName, M.Mark FROM StudentDetails E LEFT JOIN StudentMark M ON E.StudentId = M.StudentId;`

Correct answer : C. `SELECT E.StudentName, M.Mark FROM StudentDetails E LEFT JOIN StudentMark M ON E.StudentId = M.StudentId;`

Explanation :

- A `LEFT JOIN` (or `LEFT OUTER JOIN`) returns all records from the left table (`StudentDetails`), along with the matched records from the right table (`StudentMark`). If there is no match in the right table (i.e., the mark record is absent), `NULL` values are returned for the right table columns, ensuring the student's details are still displayed.
- Standard SQL join syntax uses the `ON` clause to specify the joining condition (`ON E.StudentId = M.StudentId`), making option C syntactically and logically correct.

Q15. Write pseudocode to initialize an array named `ages` with values `[25, 30, 22, 28, 35]` and print the second element of the array.

A.

```
ARRAY ages[5] = [25, 30, 22, 28, 35]
```

```
PRINT ages[2]
```

B.

```
ARRAY ages[5]
```

```
ages[1] = 25
```

```
ages[2] = 30
```

```
ages[3] = 22
```

```
ages[4] = 28
```

ages[5] = 35

PRINT ages[2]

C.

ARRAY ages[5] = [25, 30, 22, 28, 35]

PRINT ages[1]

Correct answer : C. ARRAY ages[5] = [25, 30, 22, 28, 35] \n PRINT ages[1]

Explanation :

- In standard programming and pseudocode conventions, array indexing is **0-based** (starting at index 0).
- The array elements correspond to indices as follows:
 - ages[0] = 25 (1st element)
 - ages[1] = 30 (2nd element)
 - ages[2] = 22 (3rd element)
- Therefore, to print the second element (30), the correct index to access is ages[1]. Initializing the array directly in one line as ARRAY ages[5] = [25, 30, 22, 28, 35] and executing PRINT ages[1] is the correct and concise pseudocode representation.

Q16. Which of the given options (as values for s) could possibly print the output as "Valid" in the following pseudocode?

```
function validCheck(string s):  
  if s matches "[a-zA-Z0-9]{6}"  
    then print "Valid"  
  else  
    then print "Not Valid"  
  endif  
end function validCheck
```

A. arun32

B. kkvarun32

C. arun\$2

D. arun@2

Correct answer : A. arun32

Explanation :

- The regular expression `[a-zA-Z0-9]{6}` checks for a string that contains exactly **6 alphanumeric characters** (letters or digits).
- **"arun32"** contains only alphanumeric characters and has a length of exactly **6** characters, satisfying the pattern and printing **"Valid"**.
- Option **B ("kkvarun32")** has **9** characters (fails length requirement).
- Options **C ("arun\$2")** and **D ("arun@2")** contain special characters (**\$** and **@**), which are not allowed by `[a-zA-Z0-9]`.

Q17. What will be the output of the depicted pseudo code?

Integer a,b,c

Set a=8, b=5, c=6

for(each c from 3 to 4)

 b=9+b

 b=(b+7)+a

 b=(b&4)+b

End for

Print a+b

A. 78

B. 69

C. 55

D. 65

Correct answer : D. 65

Explanation :

- **Initial values:** a = 8, b = 5, c = 6
- The loop runs for **c = 3** and **c = 4** (2 iterations):
- **Iteration 1 (c = 3):**
 - b = 9 + 5 = 14
 - b = (14 + 7) + 8 = 29
 - b = (29 & 4) + 29
 - 29 in binary is 11101 and 4 is 00100, so 29 & 4 = 4.
 - b = 4 + 29 = 33
- **Iteration 2 (c = 4):**
 - b = 9 + 33 = 42

- $b = (42 + 7) + 8 = 57$
- $b = (57 \& 4) + 57$
 - 57 in binary is 111001 and 4 is 000100, so $57 \& 4 = 0$.
 - $b = 0 + 57 = 57$
- **Final Output:**
 - Print $a + b = 8 + 57 = 65$

Q18. What will be the output of the following pseudo code for $a=0$, $b=2$, $c=7$?

Integer funn(Integer a, Integer b, Integer c)

if($(a+c+b)<(3+b-a)$)

$a=b+c$

if($(b+c+a)<(9+a+6)$)

$a=(b+a)+b$

Else

$b=(3+4)+a$

End if

$c=(b+c)+c$

End if

Print $a+b+c$

- A. 6
- B. 9
- C. 13
- D. 27

Correct answer : B. 9

Explanation :

1. Initial values are given as **$a = 0$** , **$b = 2$** , and **$c = 7$** .
2. Evaluate the outer if condition:
 - Left side: $(a + c + b) = 0 + 7 + 2 = 9$
 - Right side: $(3 + b - a) = 3 + 2 - 0 = 5$

- Condition: $9 < 5 \rightarrow$ **False**
- 3. Since the condition is false, the entire block inside the outer if is skipped, leaving the values of a, b, and c unchanged.
- 4. Print $a + b + c$ computes $0 + 2 + 7 = 9$

Q19. What will be the output of the following pseudo code?

Integer p,q,r

Set $p=4$, $q=8$, $r=6$

if($(p+r)>(q-p)$)

$q=(q+7)+q$

 if($(4+q^p)<(7+p+r)$)

$p=(q^p)^r$

 Else

$r=5+q$

 End if

End if

Print $p+q+r$

A. 50

B. 59

C. 55

D. 63

Correct answer : C. 55

Explanation: Initially, $p=4$, $q=8$, $r=6$. The first condition is true, so $q=(8+7)+8=23$. The nested condition $4+(23^4)<7+4+6$ is false because $23^4=19$, so $r=5+23=28$. Therefore, $p+q+r = 4+23+28 = 55$.

Q20. What will be the output of the following pseudo code for a=1, b=6, c=7?

Integer funn(Integer a, Integer b, Integer c)

if((a+8+c)<(6+a))

 a=(a+b)+b

 if((a+b+c)<(b+c+a))

 a=c+c

 Else

 c=(c+a)+b

 End if

 c=7+b

End if

return a+b+c

A. 14

B. 20

C. 9

D. 24

Correct answer : A. 14

Explanation: For a=1, b=6, c=7, the first condition $(a+8+c) < (6+a)$ becomes $16 < 7$, which is **false**. Therefore, none of the if statements execute, and the function returns $a+b+c = 1+6+7 = 14$.